Tailoring Machine Learning Weather Predictions for Local Impacts

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Advancements in ML Weather Forecasting have made Tailored Forecasting Models Possible

- Traditional Physics Based Weather Forecasting Models Are Expensive and Slow to Run
- New generation of data-driven ML weather models are competitive in accuracy and can be run on a laptop
- New ML models are still very expensive to train and try to jointly predict the entire globe at once.
- If we instead just focus on weather variables most relevant to people in a specific region, can we improve accuracy?

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Goal: Demonstrate that is is possible to meaningfully improve the accuracy of relevant weather variables using a small amount of compute power and data.

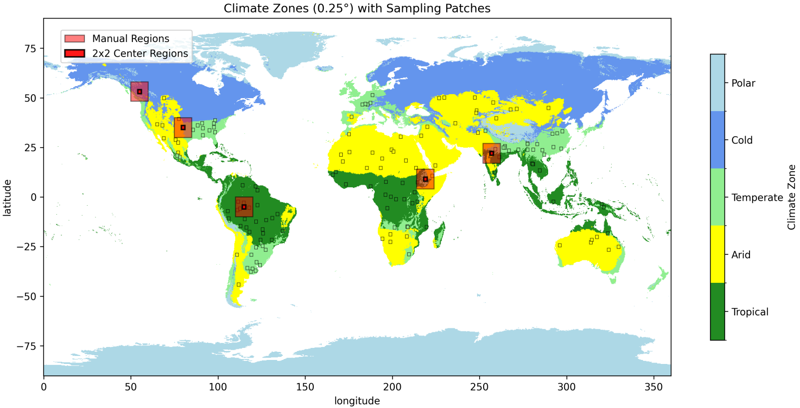
Research Questions and Early Results

- Is it possible to improve forecast accuracy using easily available off the shelf tools?
- Can NWP and ML weather models can be corrected in the same way?
- We train a simple neural network to correct both a data driven ML forecast (Pangu) and a traditional NWP forecast (IFS HRES) we find success predicting 10m wind speed but not so much 2m temperature.

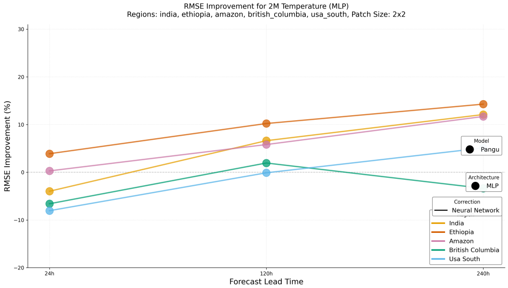
Key Contribution: Apply neural network post-processing to both NWP and ML weather models, showing that it is possible to improve the accuracy of both with perhaps some caveats.

Start by Manually Looking at Four Regions of Interest

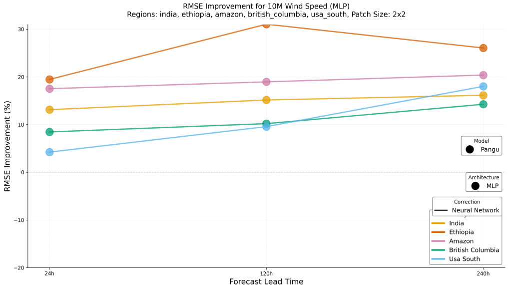
- Choose a handful of regions with a variety of climates and land types.



Do Better with Temperature: 2x2



(a) 2m Temperature: Pangu



(b) Wind Speed: Pangu

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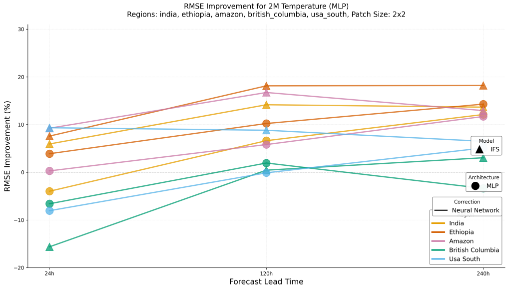

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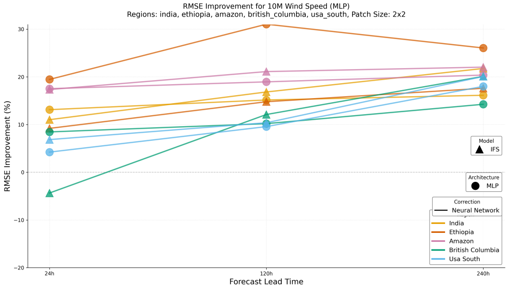

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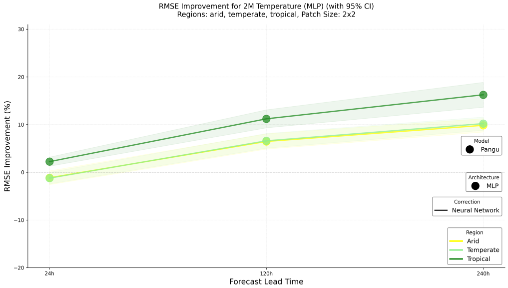
Climate Zones



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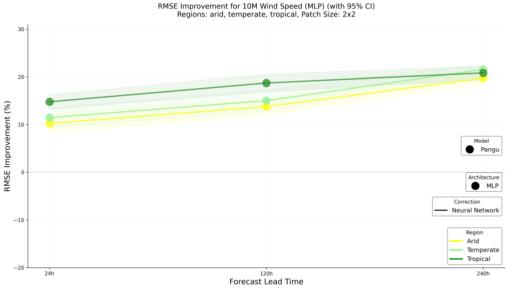

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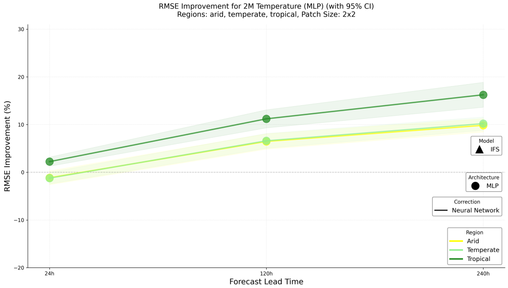

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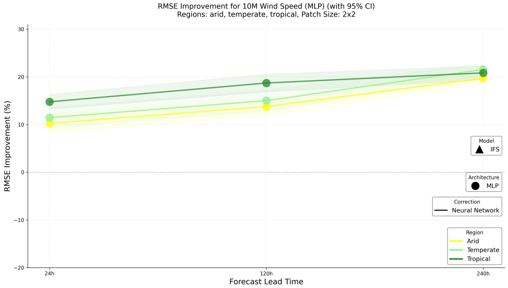

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